ALEX ANTONY

Software Engineer | Python Developer | AI/ML Engineer | Full-Stack Developer
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PROFESSIONAL SUMMARY

MCA graduate and Software Engineer with strong Python skills and hands-on experience building production-ready AI and full-stack applications. Built a bilingual Retrieval-Augmented Generation (RAG) mobile learning app with spaced repetition (FastAPI, React Native, ChromaDB), a YOLOv8-based real-time traffic management system, an AI-powered Voice Interview Coach (LangChain, FastAPI, React), and a Django-based service management platform. Currently a Software Intern at Fuselage Innovations. Skilled in the Software Development Life Cycle (SDLC), Object-Oriented Programming (OOP), RESTful API design, Data Structures and Algorithms (DSA), and full-stack development. Strong mathematical foundation (B.Sc. Mathematics). Seeking an entry-level Software Engineer role in AI engineering, Python/Django, or full-stack development.

TECHNICAL SKILLS

Programming Languages: Python (primary), JavaScript, TypeScript, Java, C++, Dart

Software Development: Object-Oriented Programming (OOP), REST API Design, SDLC, Agile, Unit Testing, Integration Testing, System Testing, Version Control (Git)

Web & Backend Frameworks: FastAPI, Django, Django REST Framework, Flask, Node.js, React.js, React Native, Flutter, HTML, CSS, Bootstrap

AI / ML & Generative AI: LangChain, Groq API, Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), Prompt Engineering, YOLOv8 (PyTorch), scikit-learn, NumPy, Pandas, Matplotlib, OpenCV

Databases: MySQL, SQLite, ChromaDB, Relational Database Design

Tools & Platforms: Git, GitHub, Linux, Docker (basic), VS Code, Cursor, Postman, Expo, Render, Vercel

Core Concepts: Data Structures & Algorithms (DSA), Problem Solving, System Design (basics), Statistics, Probability, Linear Algebra

EXPERIENCE

Software Intern | Fuselage Innovations, Kalamassery, Kochi | Sep 2026 – Present

- Working as a Software Intern at a drone and agri-tech hardware company, developing cross-platform applications using Flutter.
- Developing a Ground Control Station (GCS) application, a Mission Planner-style software for drone mission planning and monitoring.

EDUCATION

Master of Computer Applications (MCA) | 2024 – 2026
College of Engineering, Chengannur – APJ Abdul Kalam Technological University | CGPA: 8.26

Bachelor of Science in Mathematics | 2020 – 2024
SB College, Changanacherry – Mahatma Gandhi University | CGPA: 7.0

PROJECTS

Second Brain – AI-Powered Learning App with RAG & Spaced Repetition | GitHub | Aug 2026

- Built a bilingual (English/Malayalam) mobile learning app using FastAPI, React Native (Expo), Groq LLM (GPT-OSS 120B), and ChromaDB for Retrieval-Augmented Generation (RAG) to deliver context-aware, personalized topic explanations with real-world analogies.
- Designed a custom spaced-repetition scheduling algorithm (SQLite-backed) implementing an SM-2-inspired interval system to optimize long-term retention.
- Implemented an active-recall self-test feature generating novel interview-style questions from prior learning history, with user-response capture and model-answer comparison.
- Engineered a dual-LLM-call prompt architecture to solve prompt-length and topic-drift issues in low-resource-language (Malayalam) generation, improving output completeness and topic fidelity.
- Organized content into user-defined folders with cross-topic semantic search via vector embeddings; diagnosed and migrated off a deprecated LLM model in a live integration.

  Tech Stack: Python, FastAPI, React Native, Expo, Groq API, ChromaDB, SQLite, Prompt Engineering, RESTful API Design

Smart Traffic Management System (MCA Final Year Project) | GitHub | Mar 2026

- Designed and implemented a real-time vehicle detection and classification system using YOLOv8, processing live video streams with 90%+ detection accuracy.
- Built a data preprocessing pipeline using OpenCV, including frame extraction, resizing, normalization, and enhancement for varying lighting and traffic conditions.
- Developed a RESTful backend using Flask with model inference and video streaming endpoints; built a React.js dashboard for live monitoring and visualization.
- Implemented an adaptive traffic signal control algorithm (RED/YELLOW/GREEN phases) driven by real-time vehicle density estimation, reducing manual intervention.
- Conducted unit, integration, and system testing; evaluated the model using Precision, Recall, F1-score, and mAP metrics.

  Tech Stack: Python, YOLOv8 (PyTorch), OpenCV, Flask, React.js, NumPy, Pandas

Engineering Task Tracker – Cross-Platform Project Management App | GitHub | Sep 2026

- Developed a mini project management system for engineering teams using Flutter and SQLite to create, organize, and track tasks and projects through a clean cross-platform interface.
- Packaged and delivered the app as a standalone Android APK and Windows executable with local SQLite storage, requiring no hosting or internet connection.
- Executed the complete SDLC in one week (requirement analysis, system design, development, testing, deployment) with full documentation: SRS, design document, test plan and report, deployment guide, and user guide.

  Tech Stack: Flutter, Dart, SQLite, SDLC, Agile

AI Interview Coach | GitHub | Jun 2026

- Built a full-stack AI voice interview simulator using Python, LangChain, Groq LLM (LLaMA 3.3 70B), FastAPI, and React.js with voice-to-voice interaction.
- Implemented continuous speech recognition (Web Speech API) with manual End Answer control, preventing auto-cutoff during natural pauses.
- Designed RESTful FastAPI endpoints for dynamic question generation across Technical, HR, and Mixed modes with Easy, Medium, and Hard difficulty levels.
- Built resume-based personalized question generation by analyzing an uploaded PDF resume for the candidate's projects, skills, and experience.
- Developed automated answer scoring (0–10) with structured feedback on strengths, improvements, and better answers, plus session progress analytics.

  Tech Stack: Python, LangChain, Groq API, FastAPI, React.js, Web Speech API, REST API

GameHub – Mobile Game Discovery App | GitHub | Apr 2026

- Built a cross-platform mobile application using React Native (Expo) with TypeScript, targeting Android and iOS from a single codebase.
- Implemented file-based routing using Expo Router and structured the project following mobile development best practices.

  Tech Stack: React Native, Expo, TypeScript, JavaScript, Expo Router

Autocare – Vehicle Service Management System | GitHub | Oct 2025

- Developed a multi-role web application (Customer, Mechanic, Manager, Admin) with authentication, authorization, and role-based access control (RBAC) using Django.
- Designed a normalized relational database schema in MySQL; implemented MVC architecture with a modular code structure.

  Tech Stack: Django, MySQL, Bootstrap, Python

CERTIFICATIONS

- SWAYAM (NPTEL) – Cloud Computing

ACTIVITIES & INTERESTS

- Actively solving DSA problems on LeetCode (arrays, strings, trees, graphs, dynamic programming) and maintaining a DSA practice repository on GitHub.
- Building personal AI projects to stay current with LLM and Generative AI (GenAI) developments.